

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IV

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Date of Assignment: 14/08/2026	Date/Time of Submission: 14/08/2026

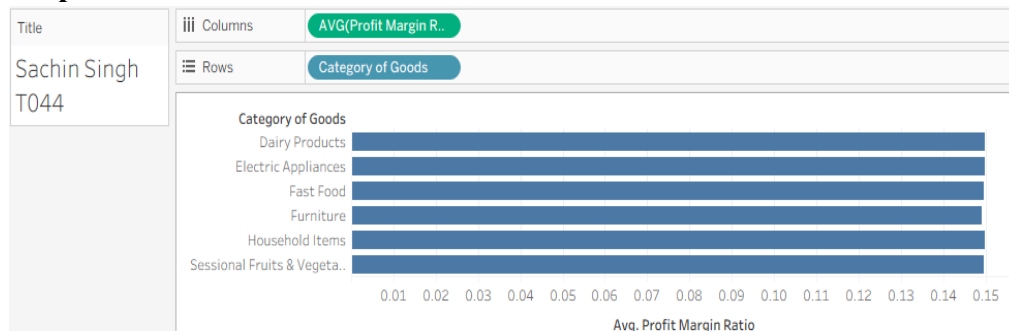
Performing Data Analysis Using Calculations in Tableau

a) Create Basic Calculated Fields Using Formulas

Steps:

1. Open store_sales_data.csv in Tableau.
2. Go to **Data Pane** → **Create Calculated Field**.
3. Name it **Profit Margin Ratio**.
4. Enter:
[Profit] / [Sales]
5. Click **OK**.
6. Add Category of Goods → Rows and Profit Margin Ratio → Columns.
7. Change the calculation to **Average**.

Output:

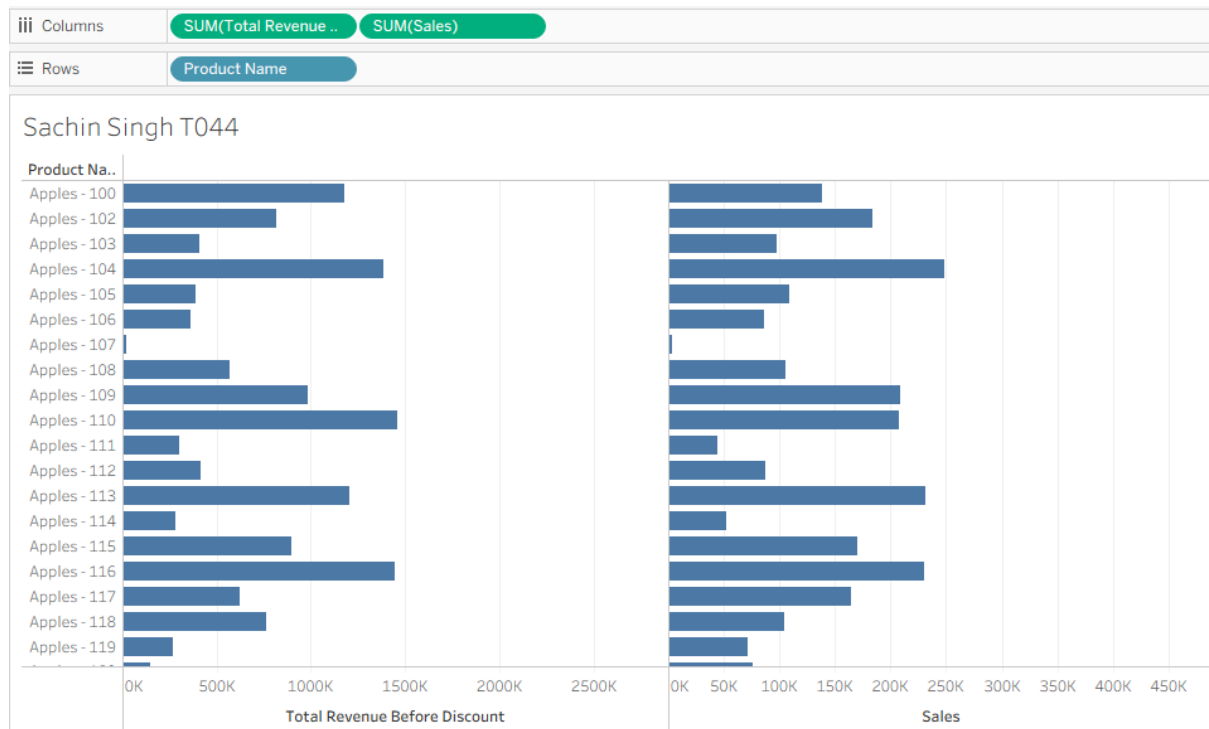


b) Perform Row-Level Calculations

Steps:

1. Create a new **Calculated Field**.
2. Name it **Total Revenue Before Discount**.
3. Enter:
[Sales] * [Quantity]
4. Click **OK**.
5. Add Product Name → Rows.
6. Add Total Revenue Before Discount → Columns.
7. Add Sales → Columns for comparison.

Output:



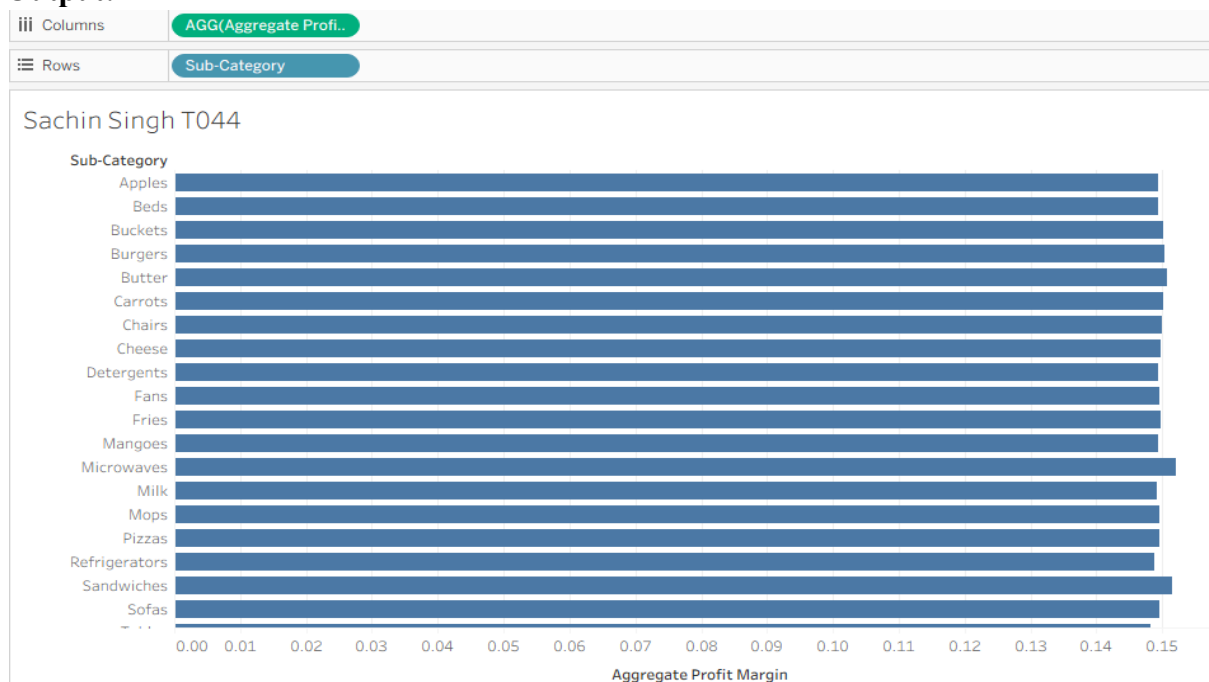
c) Perform Aggregate-Level Calculations

Steps:

1. Create a new **Calculated Field**.
2. Name it **Aggregate Profit Margin**.
3. Enter:

$$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$$
4. Click **OK**.
5. Add Sub-Category → Rows.
6. Add Aggregate Profit Margin → Columns.

Output:

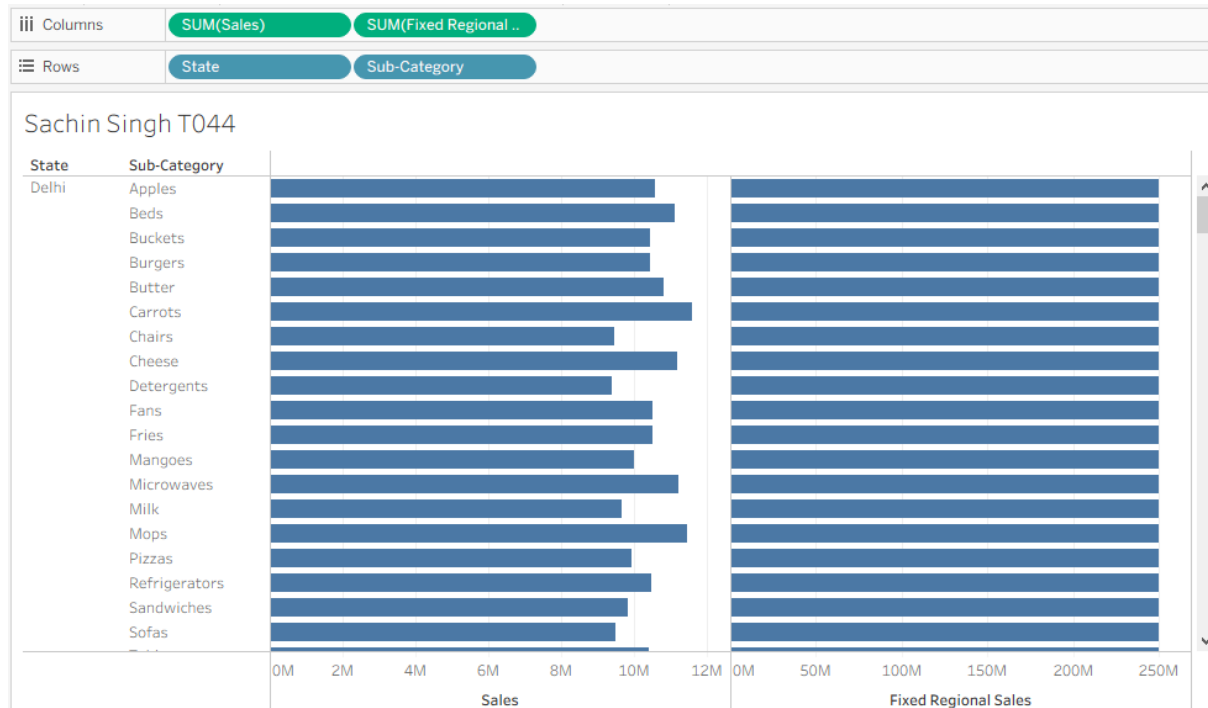


d) Implement FIXED Level of Detail (LOD) Calculations

Steps:

1. Create a new **Calculated Field**.
2. Name it **Fixed Regional Sales**.
3. Enter:
 $\{ \text{FIXED } [\text{State}] : \text{SUM}([\text{Sales}]) \}$
4. Click **OK**.
5. Add State → Rows.
6. Add Sub-Category → Rows.
7. Add Sales and Fixed Regional Sales → Columns.

Output: State-wise Fixed Regional Sales compared with Sub-Category Sales.

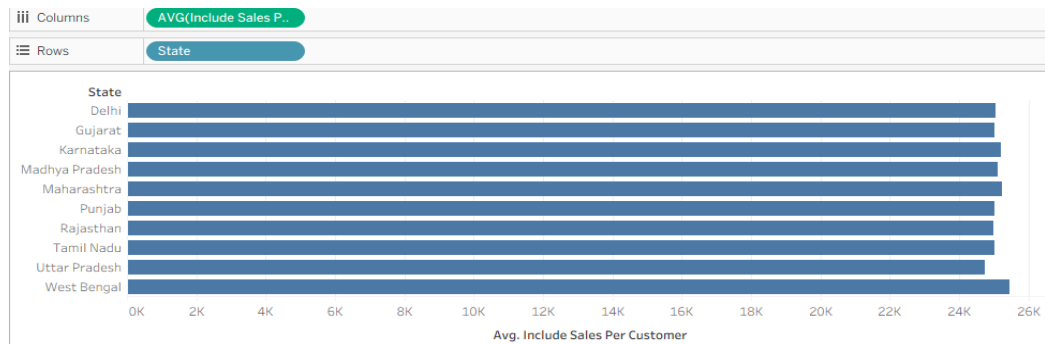


e) Use INCLUDE and EXCLUDE LOD Expressions

Steps:

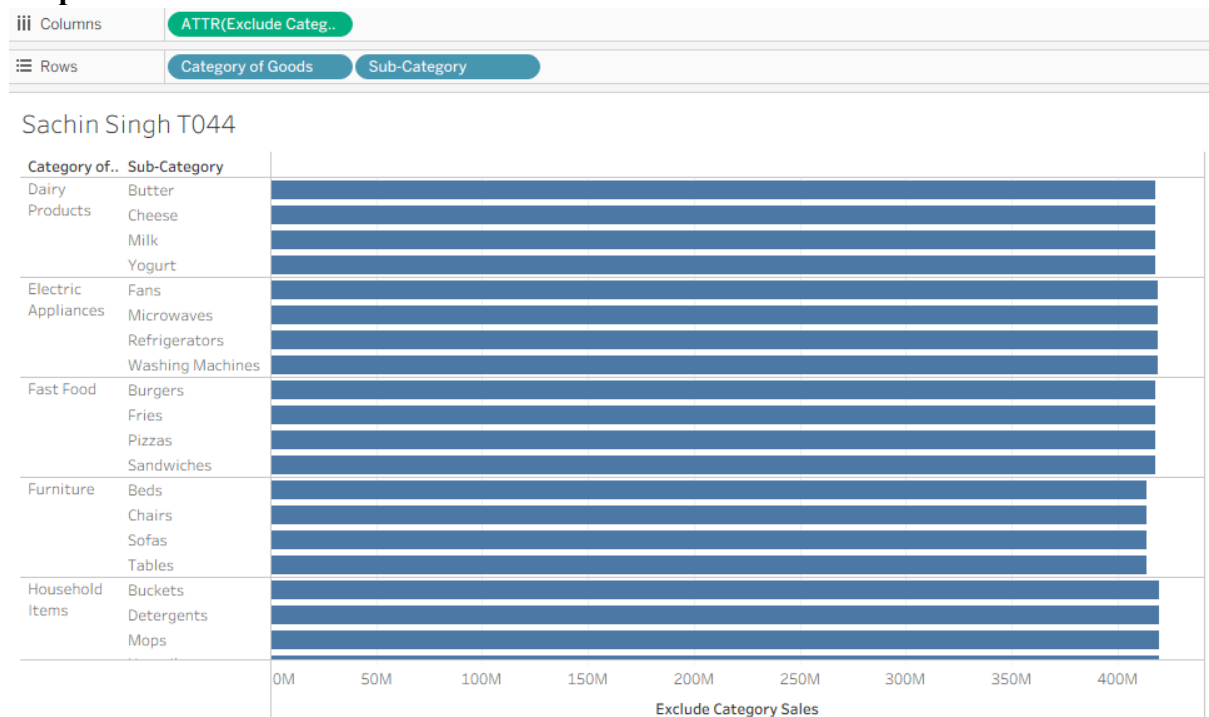
INCLUDE LOD:

1. Create a calculated field named **Include Sales per Customer**.
2. Enter:
 $\{ \text{INCLUDE } [\text{Customer ID}] : \text{SUM}([\text{Sales}]) \}$
3. Add State → Rows.
4. Add Include Sales per Customer → Columns.
5. Change the aggregation to **Average**.

**EXCLUDE LOD:**

1. Create a calculated field named **Exclude Category Sales**.
2. Enter:

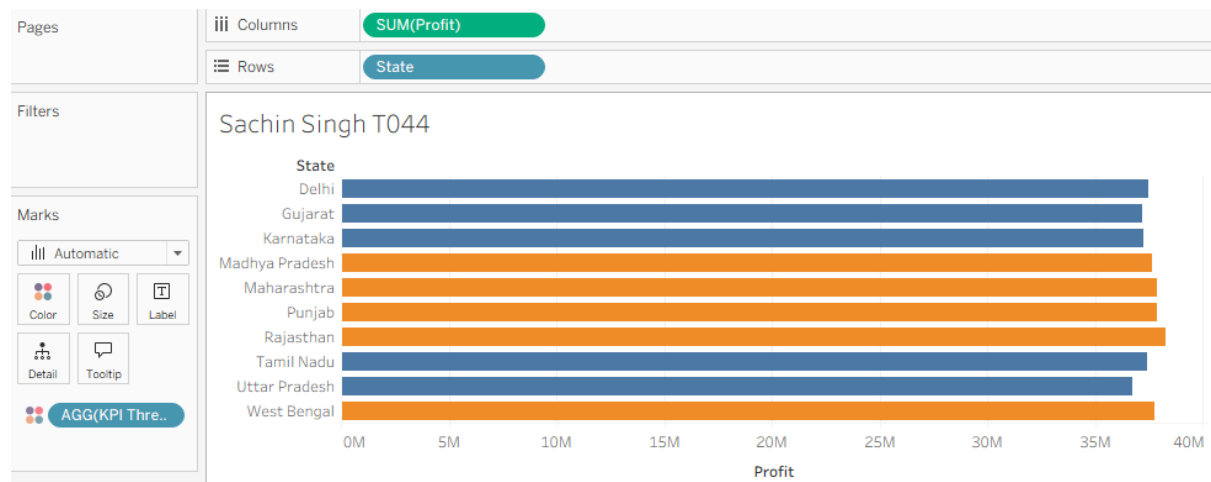
$$\{ \text{EXCLUDE [Sub-Category]} : \text{SUM}([\text{Sales}]) \}$$
3. Add Category of Goods and Sub-Category → Rows.
4. Add Exclude Category Sales → Columns.

Output:**f) Create and Use Parameters for Dynamic Analysis****Steps:**

1. Go to **Data Pane** → **Create Parameter**.
2. Name it **Select Profit Threshold**.
3. Set Data Type to **Float/Integer**.
4. Set Current Value to **500**.
5. Create a calculated field named **KPI Threshold Met?**
6. Enter:

$$\text{IF SUM}([\text{Profit}]) \geq [\text{Select Profit Threshold}] \text{ THEN "Target Achieved" ELSE "Below Target" END}$$
7. Add State → Rows and Profit → Columns.
8. Add KPI Threshold Met? → Color.
9. Select **Show Parameter Control**.

Output: Interactive Profit Threshold visualization showing **Target Achieved** and **Below Target** dynamically.



g) Apply Calculations to Fix Data Issues and Enhance Insights

Steps:

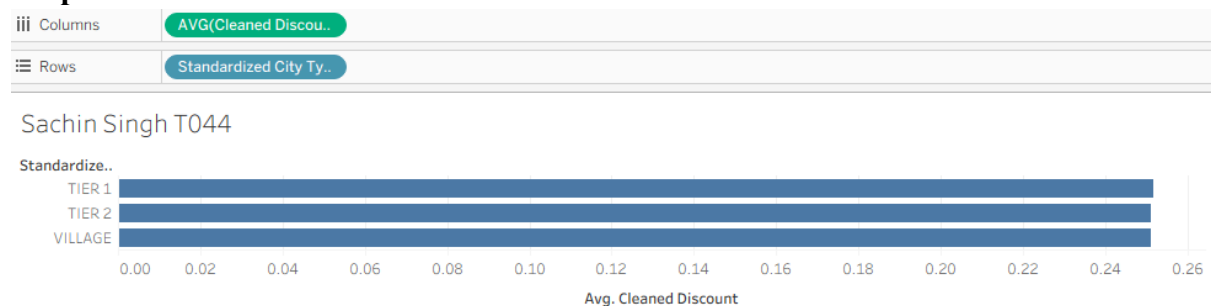
Handling Null Values:

1. Create a calculated field named **Cleaned Discount**.
2. Enter:
IFNULL([Discount], 0)
3. Click **OK**.

Standardizing Text:

1. Create a calculated field named **Standardized City Type**.
2. Enter:
UPPER([City Type])
3. Click **OK**.
4. Add Standardized City Type → Rows.
5. Add Cleaned Discount → Columns.
6. Change aggregation to **Average**.

Output:



Result

The practical was successfully performed in Tableau using calculated fields, row-level and aggregate calculations, FIXED/INCLUDE/EXCLUDE LOD expressions, parameters, and data-cleaning functions.